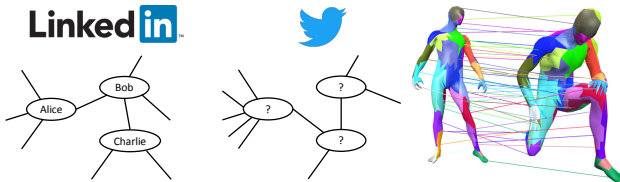


Graph Matching for Stochastic Block Models

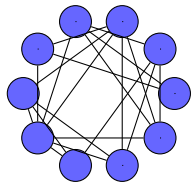
Jian Ding, Peking University

with Guanyi Chen, Shuyang Gong, Zhangsong Li

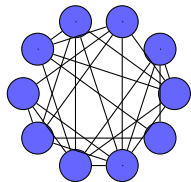
CSIAM 24th Annual Meeting



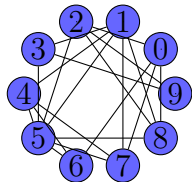
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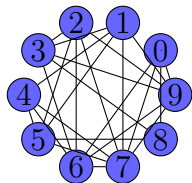
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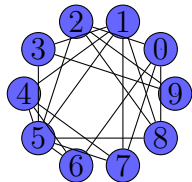
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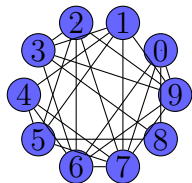
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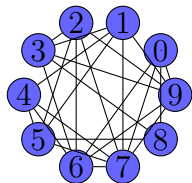
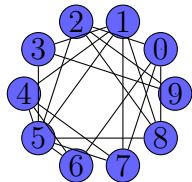
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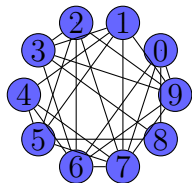
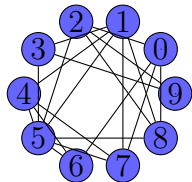


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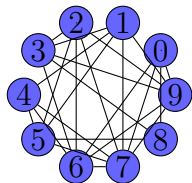
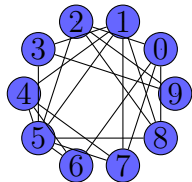
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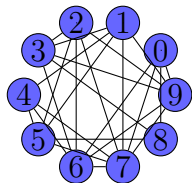
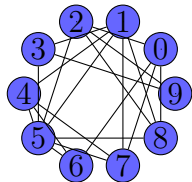
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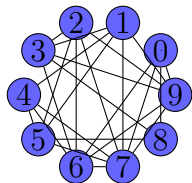
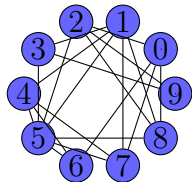
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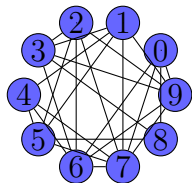
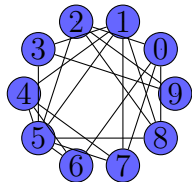
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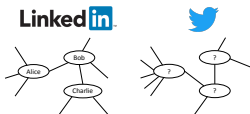
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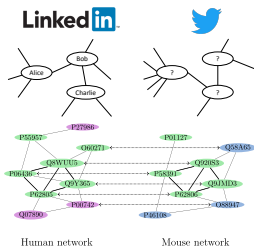
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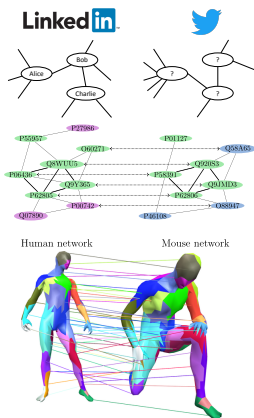
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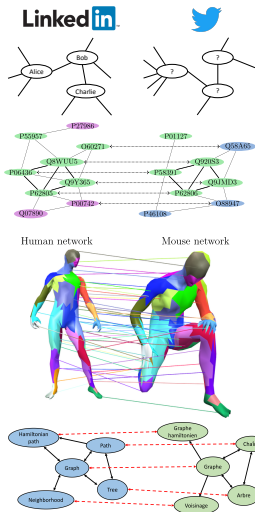
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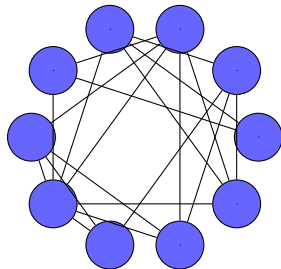
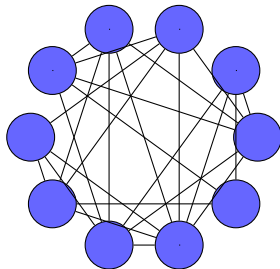
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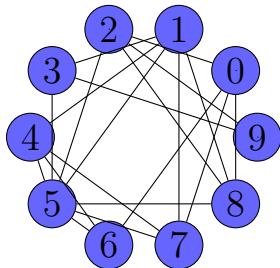
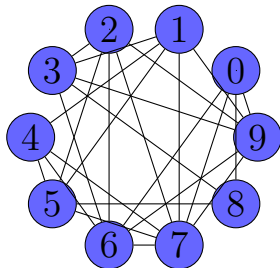


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Mathematical formulation of graph matching

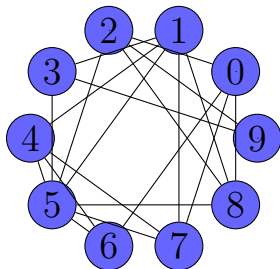
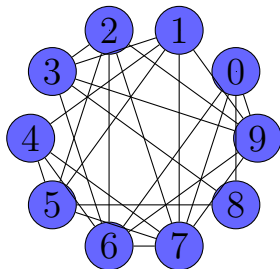


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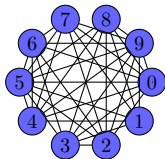
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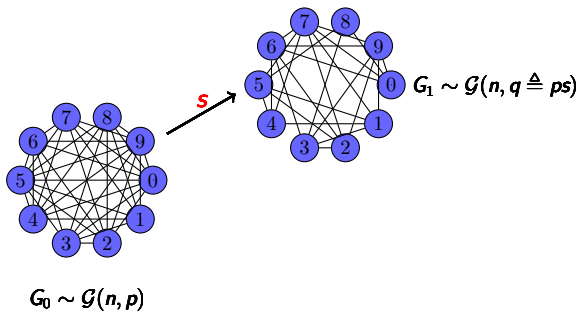
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- Since graph matching is a **hard** optimization problem (**NP-hard**), we seek help from **randomness**.

An idealized model: Correlated Erdős-Rényi graphs model

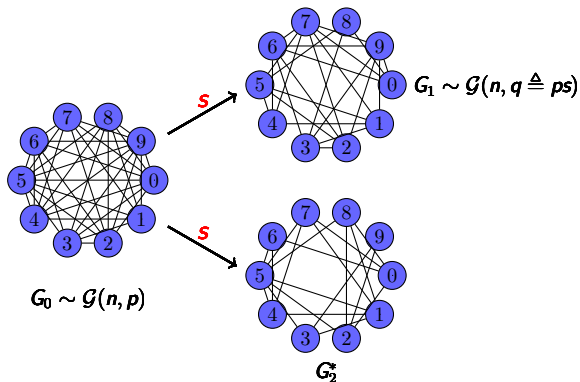


$$G_0 \sim \mathcal{G}(n, p)$$

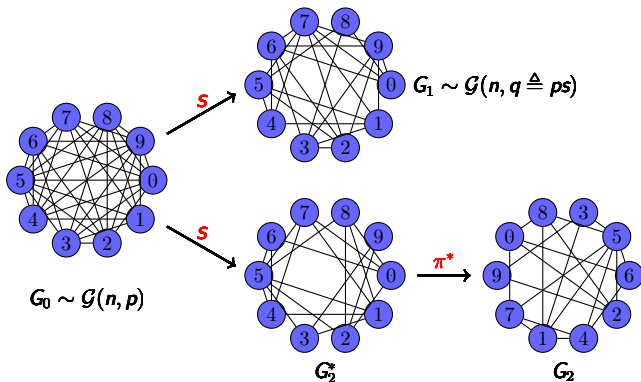
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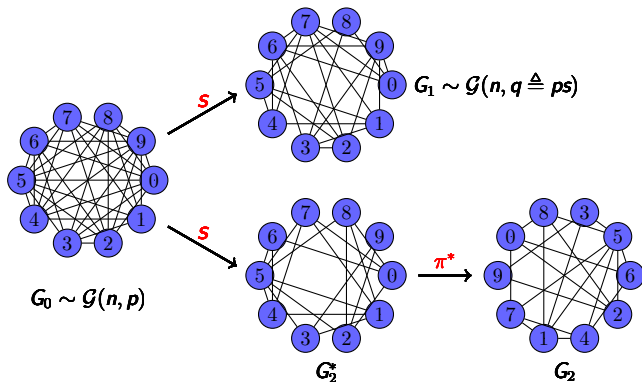
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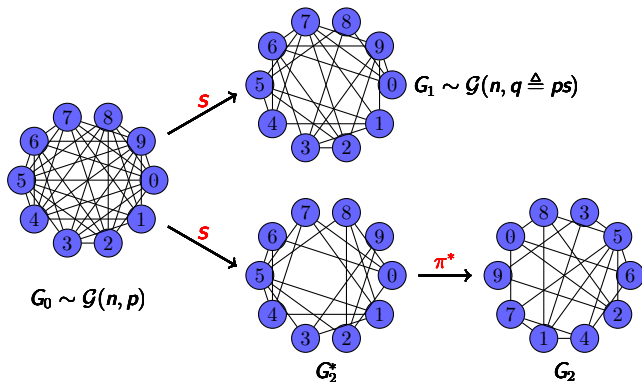


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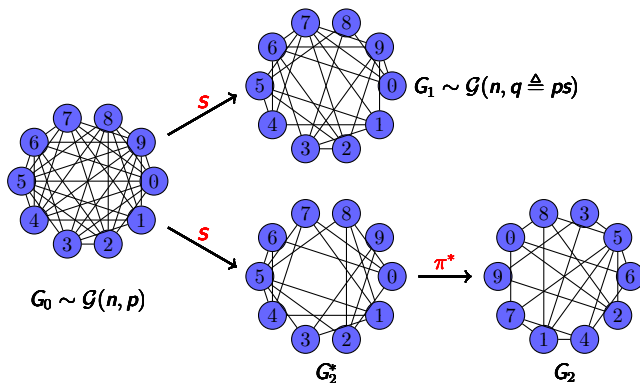
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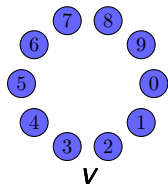


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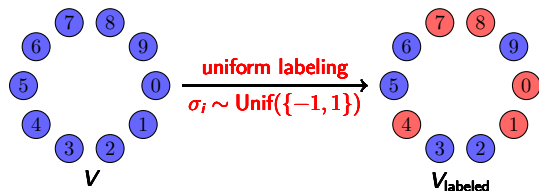
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Disadvantage: almost all realistic networks are not Erdős-Rényi.

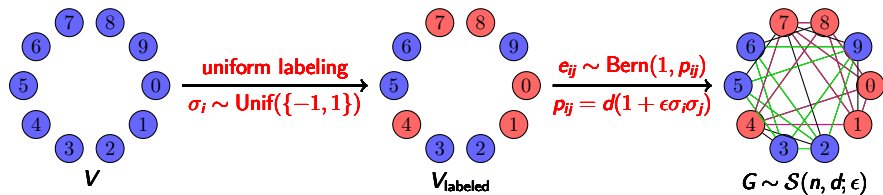
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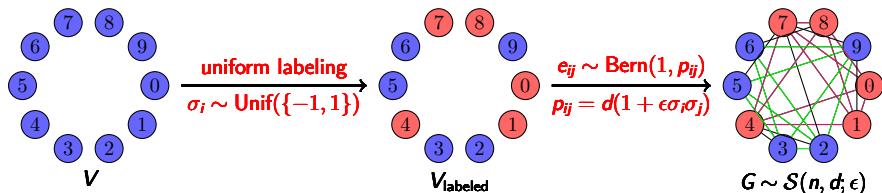
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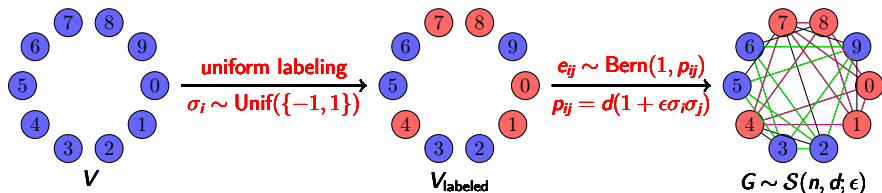


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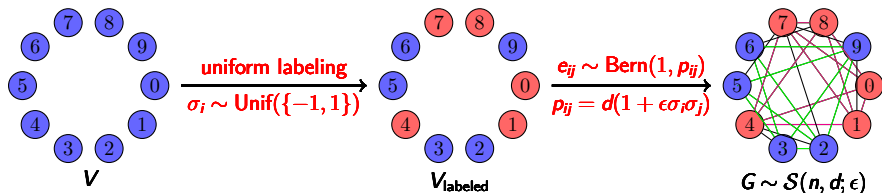
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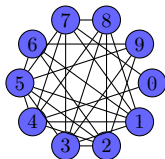
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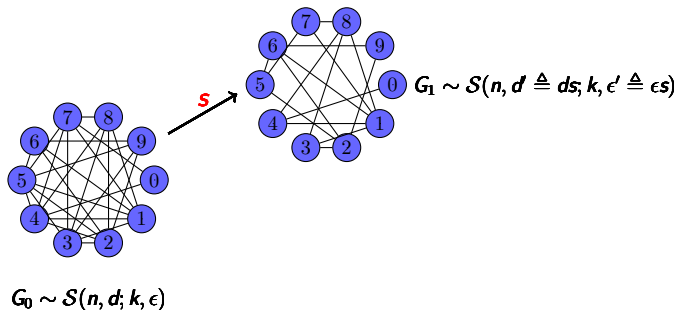
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- Generalization to k communities (denoted by $\mathcal{S}(n, d; k, \epsilon)$):
 - $\sigma_i \sim \text{Unif}(\{1, 2, \dots, k\})$;
 - $p_{ij} = d(1 + \epsilon \omega(\sigma_i, \sigma_j))$ where $\omega(\sigma_i, \sigma_j) = -1 + k \cdot \mathbf{1}_{\{\sigma_i = \sigma_j\}}$.

Correlated stochastic block model

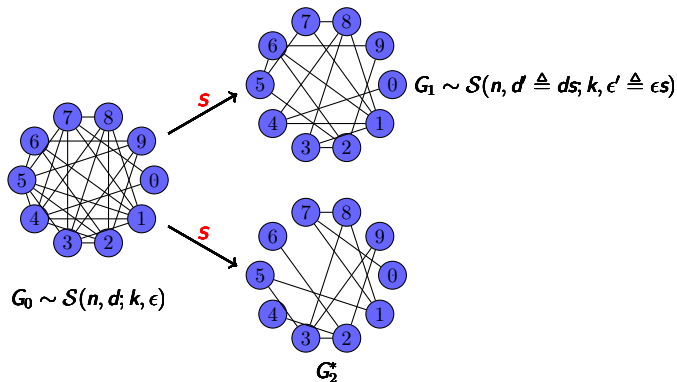


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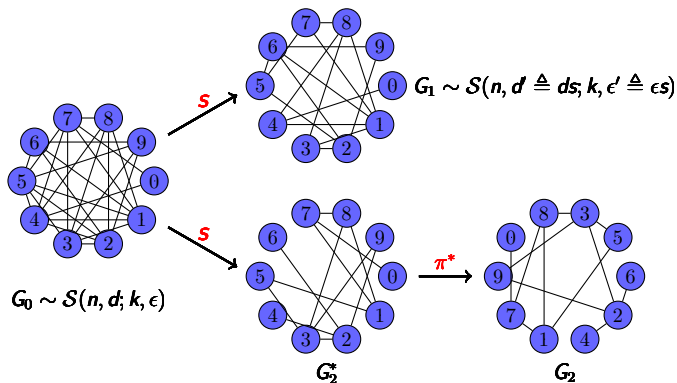
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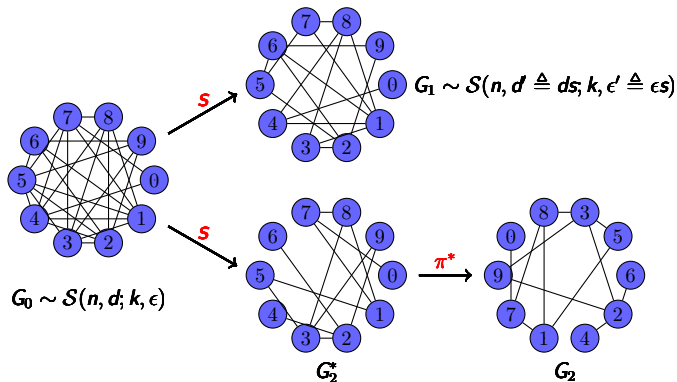
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Correlated stochastic block model



- Generated by independently subsampling $G_0 \sim \mathcal{S}(n, d; k, \epsilon)$ with probability s .
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Three thresholds: *detection*, *exact recovery*, *partial recovery*.

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- **D.-Du'23, Du25+**: Exact **detection** and **partial recovery** threshold in the **non-dense** regime, via **densest subgraph**.

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- **D.-Li'22+, 23+**: poly-time **iterative** algorithm when correlation is **non-vanishing** and average degree grows polynomially.

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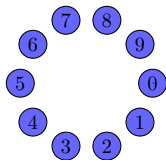
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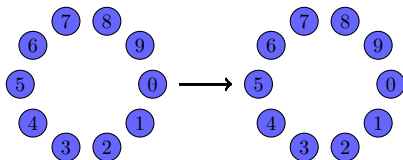
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- **Information-computation gap**: a **major** challenge in many random combinatorial optimization and constraint satisfaction problems!

Finding communities in stochastic block model



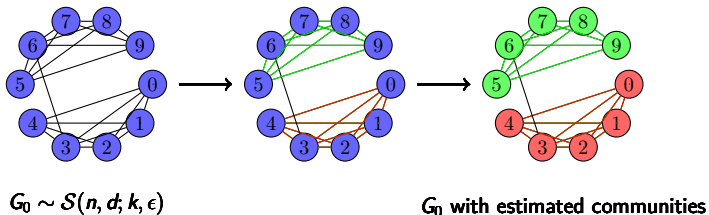
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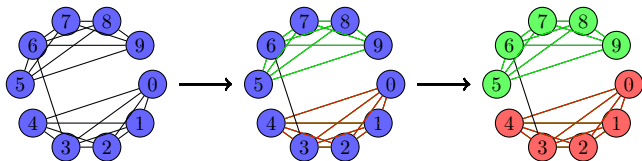


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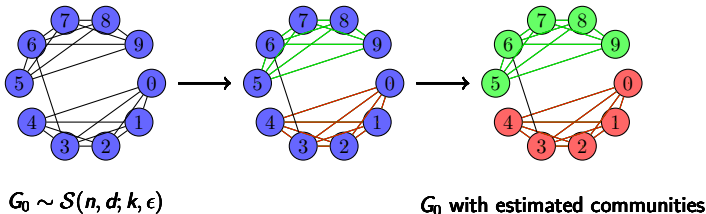


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G_0 with estimated communities

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- **Decelle-Krzakala-Moore-Zdeborová'11:**
 - Computational transition (under low-degree conjecture) around Kesten-Stigum threshold, i.e., $\lambda \epsilon^2 = 1$ **Hopkins-Steurer'16**.
 - Presumably information-computation gap if and only if $k \geq 5$ (**Mossel-Neeman-Sly'16**, **Massoulié'16** for $k = 2$ and **Mossel-Sly-Sohn'23,24** for general k).

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- Is there any case that we can break the algorithmic barrier in Erdős-Rényi model?

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Using a reduction technique in **Li'25+**, indicates in the subcritical regime (i.e., when community signal cannot be extracted from a single graph) the detection threshold between correlated/independent block models is still Otter's threshold.

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Chen-D.-Gong-Li'25+: Extends the above result to all supercritical block models with two symmetric communities (i.e., $\lambda\epsilon^2 > 1$).

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- Bridging **what is wanted** with **what is possible**.

Reference: all mentioned works available on arXiv.